Machine – Relays

Menu: Machine > Relays

Assigns relay functions to each relay output on the module. There are **16 relays per module** and up to **8 modules**.

Control	Action
Reset (circle)	Reset all relays to default
Renumber (+)	Renumber sections sequentially
Relay Number	The relay output number
Relay Type	The behavior of the relay (see below)
Number	For Switch type: switch number. For Section/Invert_Section type: section number.
Cancel	Discard changes
Save	Save settings

Application Control

Spraying occurs when:

- Master is ON
- A section is enabled
- In Auto, the guidance system allows application

Relay Types

Type	Description
Section	Controls an individual section. In Manual, the switch turns the section on or off. In Auto, the guidance system controls the section, but the switch can disable it.
FlowMaster	Controls overall product flow. In Manual, it follows the Master switch. In Auto, it is on only when the Master switch is on and the machine is moving. Use this to control overall product flow through an upstream valve or common shutoff.
Invert_FlowMaster	Opposite of the FlowMaster output — on when the master switch is off or (in Auto) the machine is stopped. Intended for valve wiring that requires an inverted signal, not as a bypass for section activity. Use **Bypass** if you need a relay that responds to whether sections are actually on.
Bypass	On when all sections are off; off when any section is on. Use this for a return or bypass valve that diverts product when no boom sections are applying.
Master	Follows the Master switch. Use this when a relay should follow the master logic itself rather than actual flow-enabled status.
Invert_Master	Opposite of the Master output (on when the Master output is off).
Slave	On when any section is on; off when all sections are off.
Invert_Section	Opposite of the Section output (on when the section is off).
Power	Always on.
Switch	Controlled directly by a switch (not affected by Master or Auto).
Hyd Up	Hydraulic raise.
Hyd Down	Hydraulic lower.
Tram Right	Tram marker right.
Tram Left	Tram marker left.
Geo Stop	Geofence stop output.
None	No function assigned to the relay.

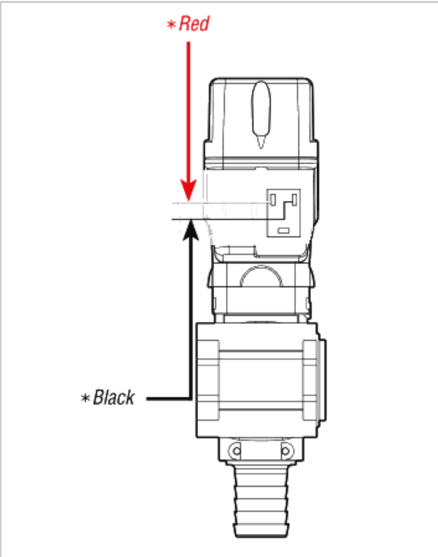
FlowMaster Valve Options

These options apply only when exactly one **FlowMaster** relay is defined on the selected module.

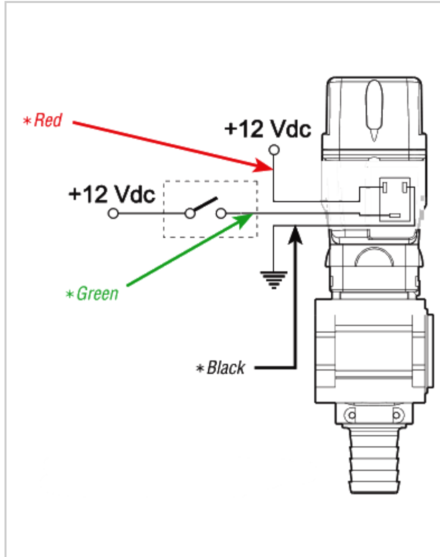
Option	Description
2-wire valve (single relay powers open and close)	One **FlowMaster** relay drives both open and close directions using the module's H-bridge output. Requires module support (e.g. RC15).
2-wire valve (2 relays: FlowMaster powers open, Invert_FlowMaster powers close)	Two relay outputs: the **FlowMaster** relay supplies power to open the valve; the **Invert_FlowMaster** relay supplies power to close it. Use this on modules without H-bridge support (e.g. RC11-2).
3-wire valve (relay signal, valve self-powered)	One **FlowMaster** relay provides a signal. The valve uses its own power supply to open or close based on that signal.

Valve Wiring Reference

The diagrams below show the wiring difference between 2-wire and 3-wire valves.



2-wire valve



3-wire valve